

DERA ISMAIL KHAN, Pakistan

WMO: 416240

Lat: 31.909N

Lon: 70.897E

Elev: 181

StdP: 99.17

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.6	(c) 5.9	(d) -4.6	(e) 2.6	(f) 15.7	(g) -1.1	(h) 3.5	(i) 12.5	(j) 5.1	(k) 14.6	(l) 3.8	(m) 14.4	(n) 0.2	(o) N/A	(p) 0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 11.8	(c) 42.9	(d) 25.9	(e) 41.5	(f) 26.1	(g) 40.1	(h) 26.6	(i) 30.6	(j) 37.7	(k) 30.0	(l) 37.1	(m) 29.5	(n) 36.4	(o) 0.4	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.0	26.2	35.7	28.3	25.1	34.9	27.7	24.3	34.3	105.7	37.9	101.9	37.2	99.3	37.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.4	3.0	2.2	2.8	45.6	1.3	1.2	1.9	46.4	1.1	47.1	0.4	47.8	-0.6	48.7
			1.6	32.0	1.3	1.3	0.6	32.9	-0.2	33.7	-0.9	34.4	-1.9	35.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	12.6	16.0	21.0	27.4	32.4	33.9	32.8	32.1	30.7	26.4	19.6	14.4	(6)
(7)		DBStd	7.96	2.35	2.57	3.58	3.29	2.64	2.57	2.22	2.15	1.98	2.66	2.23	2.47	(7)
(8)		HDD10.0	9	6	0	0	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD18.3	403	178	74	15	0	0	0	0	0	0	0	12	124	(9)
(10)		CDD10.0	5470	86	167	339	523	694	716	707	684	620	507	287	138	(10)
(11)		CDD18.3	2822	0	8	96	273	436	466	449	426	370	249	49	1	(11)
(12)		CDH23.3	39401	4	76	837	3226	6629	7427	6921	6264	5037	2562	386	31	(12)
(13)	CDH26.7	23986	0	8	294	1789	4343	5076	4474	3844	2845	1244	68	1	(13)	
(14)	Wind	WSAvg	0.5	0.3	0.6	0.5	0.6	0.7	0.6	0.5	0.4	0.3	0.2	0.3	(14)	
(15)	Precipitation	PrecAvg	355	11	25	35	26	9	28	75	85	41	10	3	6	(15)
(16)		PrecMax	744	62	119	103	74	38	98	223	368	125	75	32	27	(16)
(17)		PrecMin	154	0	0	2	0	0	0	3	5	0	0	0	0	(17)
(18)		PrecStd	138	14	27	26	23	9	26	62	84	35	17	7	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	27.8	34.5	41.2	44.8	45.0	42.5	40.3	39.2	37.0	30.2	26.0	(19)
MCWB			14.8	17.8	22.3	25.7	24.2	26.9	28.7	30.4	29.1	26.0	20.4	16.9	(20)	
2%		DB	22.0	25.5	32.1	38.7	42.9	43.3	40.4	39.1	38.0	35.6	28.6	24.1	(21)	
		MCWB	14.1	16.9	21.9	23.6	24.4	26.4	29.5	29.6	27.9	25.0	19.5	15.6	(22)	
5%		DB	20.3	24.0	29.9	36.9	41.3	41.9	39.1	38.0	36.9	34.3	27.1	22.6	(23)	
		MCWB	13.1	15.8	20.9	22.5	24.4	26.5	29.3	29.3	27.1	24.1	18.8	15.0	(24)	
10%		DB	18.8	22.2	28.0	34.8	39.8	40.5	37.9	36.9	35.9	32.7	25.7	20.8	(25)	
		MCWB	12.4	15.2	19.6	22.1	24.1	26.5	29.2	29.0	26.8	23.0	18.2	14.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	19.6	24.4	27.2	28.1	30.5	31.5	32.2	29.9	27.4	22.3	18.7	(27)
MCD B			21.9	25.5	32.3	37.3	40.0	41.2	38.6	38.2	37.3	35.3	27.9	23.8	(28)	
2%		WB	15.0	18.3	23.0	25.7	26.9	29.6	30.7	30.6	29.2	26.3	21.0	17.1	(29)	
		MCD B	20.6	23.8	30.2	36.6	39.1	39.5	37.6	37.2	36.2	33.9	26.9	22.6	(30)	
5%		WB	14.1	17.2	21.7	24.3	26.0	28.8	30.0	29.9	28.4	25.3	19.9	15.9	(31)	
		MCD B	19.2	22.4	28.5	34.3	38.4	38.2	36.9	36.4	35.0	32.5	25.4	21.2	(32)	
10%		WB	13.2	16.0	20.5	23.0	25.2	28.0	29.5	29.4	27.6	24.3	18.9	14.7	(33)	
		MCD B	17.7	20.9	26.8	32.2	37.0	36.7	36.4	35.5	33.9	30.8	24.3	19.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	11.1	10.8	12.8	13.4	11.8	9.2	8.8	9.9	12.1	12.5	12.6	(35)
MCDBR			14.4	14.1	13.4	15.4	15.5	14.0	10.8	10.2	11.0	12.7	13.9	15.2	(36)	
MCWBR			8.4	7.4	6.1	5.9	5.2	4.9	3.7	3.8	3.9	5.0	7.2	9.3	(37)	
5% WB		MCD BR	13.0	12.1	12.8	14.7	14.3	11.6	9.7	9.5	10.1	12.0	12.3	14.1	(38)	
		MCWBR	8.3	6.8	6.2	6.5	5.6	4.7	4.1	4.2	4.2	5.2	7.3	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.451	0.494	0.516	0.563	0.607	0.702	0.796	0.743	0.632	0.610	0.566	0.476	(40)	
taud		1.929	1.875	1.853	1.787	1.705	1.565	1.453	1.538	1.688	1.650	1.693	1.866	(41)		
Ebn at Noon		732	745	766	749	720	652	590	613	662	629	608	675	(42)		
Edh at Noon		155	180	197	219	240	275	306	278	230	223	196	157	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.12	3.91	5.13	6.32	7.03	6.84	5.95	5.66	5.39	4.60	3.50	3.06	(44)	
(45)		RadStd	0.23	0.38	0.36	0.29	0.22	0.29	0.25	0.32	0.27	0.18	0.20	0.12	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page